MIL-G-558B
21 August 1969
SUPERSEDING
MIL-G-558A
31 December 1958

MILITARY SPECIFICATION

GUANIDINE CARBONATE, TECHNICAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one technical grade of guanidine carbonate [$(H_2N \cdot CNH \cdot NH_2)_2 \cdot H_2CO_3$].

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-35 - Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Guanidine carbonate shall conform to the chemical characteristics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Purity	96.0	----	4.2.4.1
Moisture	----	0.50	4.2.4.2
Ash	----	0.20	4.2.4.3
Sulfate (as SO_4)	----	0.25	4.2.4.4
Insoluble matter	----	0.20	4.2.4.5
Chlorides	----	To pass test	4.2.4.6

3.2 Granulation. No more than 5.0 percent by weight of guanidine carbonate shall be retained on a 420-micron (No. 40) sieve when tested as specified in 4.2.4.7.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the guanidine carbonate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified.
(b) Container not as specified.
(c) Container closure not as specified.
(d) Container damaged or leaking.
(e) Marking incorrect, missing, or illegible.

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Purity. Determine the purity of the specimen by the total nitrogen method specified in (a) and by the titration method specified in (b). The percent purity shall be taken as the lower of the values calculated by the two methods.

(a) Purity by total nitrogen. Heat approximately 0.5 gram (g) of the specimen in an oven at 100° to 105° C to constant weight and cool to room temperature in a desiccator. Weigh to the nearest 0.1 milligram (mg) 0.11 to 0.14 g of the dried specimen and quantitatively transfer to a 500- or 800-milliliter (ml) Kjeldahl flask using the minimum amount of water necessary. Add 10 g of powdered potassium sulfate or anhydrous sodium sulfate and 0.5 g of powdered cupric sulfate or 0.3 g of selenium, or a suitable combination of the catalyst. (A quantity of 0.78 g of mercuric oxide or an equivalent amount of mercury may be used to replace the copper sulfate or selenium, but in this case sodium sulfide must be added later to remove the mercury before distillation, thus preventing the formation of mercury-amido compounds which are not completely decomposed by sodium hydroxide.) Next, add 20 ml of concentrated sulfuric acid, pouring it slowly down the neck so that any of the specimen adhering to the neck of the flask is washed down. Incline the flask at an angle of approximately

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45° and gently heat over a free flame or electrically until frothing has ceased. Increase the heat until the acid boils briskly and continue heating until the liquid has been clear green in color, or nearly colorless, for 30 minutes. After cooling, cautiously add 150 ml of water, cool again, and then add 100 ml of 30-percent soidum hydroxide solution down the side of the flask so that it does not mix all at once with the acid solution. Add a few pieces of granulated zinc and immediately connect the flask to the Kjeldahl distillation appraratus by means of a rubber stopper. (A preferred procedure is to add the alkali through a small separatory funnel passing through a second hole in the stopper. No loss of ammonia can result with this method.) Mix the layers carefully by rotating the flask, then distill 150 to 200 ml of the solution into 50 ml of 0.1N hydrochloric acid containing a few drops of methyl red indicator. Titrate the excess acid with 0.1N sodium hydroxide solution. Calculate the percent guanidine carbonate as follows:

$$\text{Percent guanidine carbonate} = \frac{3.004\ (AB - CD)}{W}$$

where: A = Milliliters of hydrochloric acid added,
B = Normality of hydrochloric acid,
C = Milliliters of sodium hydroxide solution used,
D = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

(b) Purity by titration. Heat approximately 1 g of the specimen in an oven at 100° to 105° C to constant weight and cool to room temperature in a desiccator. Weigh to the nearest 0.1 mg 0.35 to 0.42 g of the dried specimen and quantitatively transfer to a 300-ml Erlenmeyer flask using about 50 ml of water. Add 50.0 ml of 0.1N hydrocholric acid, boil to expel carbon dioxide, cool, and titrate the excess acid with 0.1N sodium hydroxide solution using methyl red as indicator. Calculate the percent guanidine carbonate as follows:

$$\text{Percent guanidine carbonate} = \frac{9.01\ (AB - CD)}{W}$$

where: A = Milliliters of hydrochloric acid added,
B = Normality of hydrochloric acid,
C = Milliliters of sodium hydroxide solution used,
D = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.2 Moisture. Weigh to the nearest milligram approximately 2 g of the specimen into a tared, low-form weighing dish. Heat in an oven at 100° to 105° C to constant weight, cool in a desiccator, and weigh to the nearest milligram. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100\ (A - B)}{W}$$

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where: A = Weight of specimen and dish before heating in grams,
B = Weight of specimen and dish after heating in grams, and
W = Weight of specimen in grams.

4.2.4.3 Ash. Weigh to the nearest milligram approximately 2 g of the specimen into a tared platinum crucible. Heat the crucible slowly by placing it on the hearth of a muffle furnace heated to dull redness. When the specimen has been carbonized completely, place the crucible in the muffle and ash at a temperature above 750° C. Cool in a desiccator and weigh to the nearest milligram. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.4 Sulfate (as SO_4). Weigh to the nearest milligram approximately 10 g of the specimen into a 400-ml beaker. Dissolve in a slight excess of concentrated hydrochloric acid, dilute to 200 ml with water, filter if necessary, and wash. Heat to boiling and then add, dropwise, 10 ml of 10-percent barium chloride solution. Digest on a hot plate for 3 hours, filter through a tared filter crucible weighed to the nearest 0.1 mg, and wash free of chlorides with hot water. Dry the crucible and precipitate in an oven at 100° to 105° C, and then in a muffle furnace at dull red heat for 1/2 hour. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent sulfate (as SO_4) as follows:

$$\text{Percent sulfate (as } SO_4\text{)} = \frac{3.43\ (A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.5 Insoluble matter. Weigh to the nearest milligram approximately 5 g of the specimen and dissolve in 250 ml of water at approximately 80° C. Filter through a tared filter crucible weighed to the nearest 0.1 mg and wash the residue several times with water having a temperature of approximately 80° C. Dry in an oven at 100° to 105° C, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100\ (A - B)}{W}$$

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where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.6 Chlorides.

(a) Standard sodium chloride solution. Dissolve 0.165 g of sodium chloride in water to make 100 ml of solution. Dilute 10 ml of the solution to 1 liter with water.

(b) Procedure. Dissolve 2.00 g of the specimen in 20 ml of water and heat to boiling. Cool, filter, and wash with water to make 20 ml of filtrate. To 15 ml of the filtrate, add concentrated nitric acid until the solution is neutral and add 2 ml in excess. Add 1 ml of 2-percent silver nitrate solution. Prepare a standard solution by adding 2 ml of standard sodium chloride solution prepared as specified in (a) to the remaining 5 ml of filtrate, diluting to 20 ml with water, neutralizing with concentrated nitric acid, adding 2 ml in excess, and adding 1 ml of 2-percent silver nitrate solution. Any turbidity produced in the specimen solution shall be no greater than that produced in the standard solution.

4.2.4.7 Granulation. Place a clean, dry, tared 420-micron (No. 40) sieve conforming to RR-S-366 on a bottom pan. Weigh to the nearest 0.1 g approximately 100 g of the specimen and transfer to the sieve. Cover and shake until the weight of the residue retained on the sieve is reduced by no more than 0.1 g by further shaking for 1 minute. Weigh the residue on the sieve and calculate the percent retained as follows:

$$\text{Percent retained} = \frac{100\ (A - B)}{W}$$

where: A = Weight of sieve and residue in grams,
B = Weight of sieve in grams, and
W = Weight of specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. A maximum quantity of 100 pounds of guanidine carbonate shall be packed in accordance with Uniform Freight Classification Rules in a laminated textile bag conforming to type III, class 6, style optional, bag number as applicable of PPP-B-35 or in a multiwall paper sack conforming to type and style optional, construction number 13 or 13X, with moisture barrier protection grade 1 of UU-S-48. The filled container shall be closed as specified in the applicable specification.

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5.2 Marking. Marking shall be in accordance with MIL-STD-129 and as follows:

KEEP AWAY FROM HEAT AND OPEN FLAME

6. NOTES

6.1 Intended use. Guanidine carbonate is intended for use in the manufacture of a component used in protective ointments.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM
Navy - MS

Preparing activity:

Army - MU(EA)

Project No. 6810-0554

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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